Practice Exam B: Answer Key and Worked Solutions

Points need units, correct sig figs and shown work. Red = common point-losers. Grade A ≈ 172+, B ≈ 154+, C ≈ 138+ (your class's Exam 1 cutoffs).

Problem 1: % ionic character (15)

(a) δ = μ/(eR). HF: (1.826 × 3.336×10⁻³⁰ C·m)/(1.602×10⁻¹⁹ C × 91.7×10⁻¹² m) = 0.415 → **41.5% ionic**. HBr: (0.827 × 3.336×10⁻³⁰)/(1.602×10⁻¹⁹ × 141.4×10⁻¹²) = 0.122 → **12.2% ionic**.
(b) ΔEN: HF 3.98 − 2.20 = 1.78 → **"ionic" by the >1.7 rule**; HBr 2.96 − 2.20 = 0.76 → **polar covalent**. But HF is only 41% ionic (< 50%), so it is really a **strongly polar covalent** bond. The ΔEN cutoff is only a rule of thumb, and H is a non-metal, so H–F isn't a metal/non-metal pair.
(c) **Arrow from F (δ⁻) to H (δ⁺)**, and from Br to H (IUPAC: − → +). The HF arrow is longer.

Problem 2: Cyanate vs. fulminate (20)

Both have 16 valence e⁻ (5 + 4 + 6 + 1).
Cyanate (C central): [N≡C–O]⁻: FC N 0, C 0, O −1 · [N=C=O]⁻: N −1, C 0, O 0 · [N–C≡O]⁻: N −2, C 0, O +1 (poor).
Fulminate (N central): [C≡N–O]⁻: C −1, N +1, O −1 · [C=N=O]⁻: C −2, N +1, O 0 · [C–N≡O]⁻: C −3, N +1, O +1 (poor).
(b) Best cyanate structure: **[N≡C–O]⁻**, a single −1 on the most EN atom (O) and zeros elsewhere. Best fulminate structure: **[C≡N–O]⁻**, which still needs three non-zero FCs (−1, +1, −1) and a negative FC on low-EN carbon. **Cyanate is more stable** (smaller FCs, charge on supporting-EN atom); fulminate's charge separation helps explain its explosive instability.
(c) Average [N≡C–O]⁻ and [N=C=O]⁻: C–N = (3 + 2)/2 = **2.5**; C–O = (1 + 2)/2 = **1.5**.
(d) C has 2 electron domains, no LP → **linear, 180°, C sp**.

Problem 3: Bond angle reasoning (10)

(a) **NO_2^+ (180°) > NO_2• (~134°) > NO_2^- (~115°)**. NO_2^+ has no non-bonding e⁻ on N (AX_2, linear). NO_2• has a single-electron domain, which needs less space than a lone pair, so it compresses the angle only slightly. NO_2^- has a full lone pair, the largest domain (tethered to one nucleus only), which squeezes the bonds most.
(b) In the trigonal bipyramid, an **axial** site has 3 neighbors at 90°; an **equatorial** site has only 2 at 90° (+ 2 at 120°) → **more room, so the big lone pair goes equatorial**. The lone pair repels the axial S–F bonds and pushes them away, bending the axial F–S–F to 173° (and the equatorial angle to 102–106° instead of 120°).

Problem 4: PCl_5 hybridization (10)

(a) **sp^3d**. Equatorial: **3s + 3p$_x$ + 3p_y → three sp^2 hybrids** (120°). Axial: **3p_z + 3d_{z^2}** (the only d orbital pointing along z).
(b) The axial pd hybrids are higher in energy, so they form weaker, less stable bonds. Axial Cl atoms also have 3 neighbors at 90° vs. 2 for equatorial, which means more repulsion. Both make the axial bonds **longer (214 vs 202 pm)**.
(c) N (n = 2) has **no 2d orbitals**: only 2s + 2p, i.e. 4 orbitals → max 4 bonds/8 e⁻. 2nd-row elements strictly obey the octet rule (N is also too small to fit 5 Cl).

Problem 5: MO theory (20)

(a) He_2: $(\sigma g1s)^2(\sigma^* u1s)^2$ → BO = ½(2 − 2) = **0**. The antibonding MO is raised more than the bonding MO is lowered, so the net effect is repulsive and the molecule doesn't exist. He_2^+•: $(\sigma g1s)^2(\sigma^* u1s)^1$ → BO = **½**, a net bonding effect → it can exist.
(b) B_2 has 6 valence e⁻. **N_2-type** (π below σg2p): $(\sigma g2s)^2(\sigma^* u2s)^2(\pi u2p)^1(\pi u2p)^1$ → **2 unpaired e⁻, paramagnetic ✓**. **O_2-type** (σg2p below π): $(\sigma g2s)^2(\sigma^* u2s)^2(\sigma g2p)^2$ → all paired, diamagnetic ×. Experiment says paramagnetic → **the N_2-type ordering is correct**. Cause: **s–p mixing** (the σg2s/σ*u2s and σg2p MOs interact, from shielding/penetration; strong for early 2nd-row elements where the 2s–2p gap is small), which pushes σg2p above πu2p.
(c) C_2 (8 e⁻): $\ldots(\pi u2p)^4$ → **BO 2, diamagnetic**. C_2^{2-} (10 e⁻): $\ldots(\pi u2p)^4(\sigma g2p)^2$ → **BO 3, diamagnetic** (K_2C_2 is diamagnetic, isoelectronic with N_2). Bond lengths: **C_2^{2-} (BO 3) < C_2 (BO 2) < C_2H_6 (BO 1)**.
(d) O_2 has 2 unpaired e⁻ in π*g2p → **S = 2 × ½ = 1**.

Problem 6: Hydrate (10)

n($MgSO_4$) = 1.221 g/120.36 g/mol = 0.010145 mol. Water lost = 2.500 − 1.221 = 1.279 g → /18.015 = 0.07100 mol. x = 0.07100/0.010145 = 7.00 → **x = 7, $MgSO_4 \cdot 7H_2O$** (Epsom salt). Mass-% H_2O = 1.279/2.500 × 100 = **51.16%**.

Problem 7: Combustion analysis + vapor density (20)

(a) C: 0.3996 g CO_2 × (12.011/44.009) = **0.1091 g C**. H: 0.1636 g H_2O × (2.016/18.015) = **0.01831 g H**. O by difference: 0.2000 − 0.1091 − 0.0183 = **0.0726 g O**. O can't be measured directly. Get it by difference!
(b) mol C = 0.009080, H = 0.01816, O = 0.004540 → ÷ 0.004540 = 2.00 : 4.00 : 1.00 → **C_2H_4O** (44.05 g/mol).
(c) $\mathcal{M}$ = dRT/P = (2.487)(0.08206)(423.15)/(0.980) = **88.1 g/mol** → k = 88.1/44.05 = 2 → **$C_4H_8O_2$** (e.g. ethyl acetate, fruity smell).
(d) **$C_4H_8O_2 + 5\ O_2 \rightarrow 4\ CO_2 + 4\ H_2O$**.

Problem 8: Concentrated HNO_3 (15)

Take 100.0 g of solution: 70.0 g HNO_3/63.012 = 1.1109 mol; 30.0 g H_2O = 1.665 mol; V = 100.0 g/1.413 g/mL = 70.77 mL.
(a) M = 1.1109/0.07077 L = **15.7 M**; b = 1.1109/0.0300 kg = **37.0 mol/kg**; χ = 1.1109/(1.1109 + 1.665) = **0.400**. Molality uses kg of SOLVENT (30.0 g), not solution.
(b) V_1 = (1.50 M)(500. mL)/(15.70 M) = **47.8 mL**.
(c) **Always add acid to water** (the dilution is very exothermic), never water to acid. Also: fume hood, goggles, gloves.

Problem 9: Mixing ionic solutions (10)

mol Na^+ = 0.200 × 0.154 = 0.0308; mol Ca^{2+} = 0.300 × 0.100 = 0.0300; mol Cl^- = 0.0308 + 2(0.0300) = 0.0908. V = 1.000 L.
(a) **[Na^+] = 0.0308 M, [Ca^{2+}] = 0.0300 M, [Cl^-] = 0.0908 M; total ions = 0.152 M**. Check electroneutrality: 0.0308 + 2(0.0300) = 0.0908 ✓. $CaCl_2$ gives 2 Cl^-!
(b) $Ag^+ + Cl^- \rightarrow AgCl(s)$: 0.0908 mol × 143.32 = **13.0 g AgCl**.

Problem 10: Antacid (20)

(a) $NaHCO_3(aq)$: Na^+, HCO_3^-, H_2O, plus a little H_2CO_3, CO_3^{2-}, OH^- (HCO_3^- equilibria). HCl(aq): H_3O^+, Cl^-, H_2O (no HCl molecules, strong acid).
(b) **Acid/base + gas-forming**: $HCO_3^-(aq) + H_3O^+(aq) \rightarrow H_2CO_3(aq) + H_2O(l)$, then $H_2CO_3(aq) \rightarrow CO_2(g) + H_2O(l)$. Net: **$HCO_3^- + H_3O^+ \rightarrow CO_2(g) + 2\ H_2O(l)$**. Pairs: H_3O^+/H_2O and H_2CO_3/HCO_3^-. Spectators Na^+, Cl^-.
(c) It can donate H^+ (acid) or accept H^+ (base): **CA = H_2CO_3, CB = CO_3^{2-}**.
(d) $NaHCO_3$ = 0.840/84.007 = 0.01000 mol; HCl = 0.0250 × 0.300 = 0.00750 mol; 1:1 → **HCl limiting**. CO_2 = 0.00750 mol; V = nRT/P = (0.00750)(0.08206)(310.15)/1.00 = **0.191 L = 191 mL**.
(e) Going down a group, the **size effect beats EN**: Cl^- is larger than F^-, so its −1 charge is spread over more volume (lower charge density) → more stable conjugate base → stronger acid. F^- is small with high charge density, so HF holds on to its proton (weak acid).

Problem 11: Redox via Lewis structures (15)

(a) $H_3C–CH_2–O–H$. Give the bond e^- to the more EN atom (split C–C): CH_3 carbon gets 3 e^- from H plus 1 from C–C → **ON = −3**. CH_2OH carbon: +2 e^- from H, loses the C–O e^- to O → **ON = −1**. O −2, H +1 (sum: −3 − 1 − 2 + 6 = 0 ✓).
(b) O_2: O 0. CO_2: C +4, O −2. H_2O: H +1, O −2.
(c) C is oxidized (−3 → +4, −1 → +4): **ethanol = reducing agent**. O is reduced (0 → −2): **O_2 = oxidizing agent**. Electrons: 7 + 5 = **12 e^- per ethanol** (check: 3 O_2 × 2 O × 2 e^- = 12 ✓).
(d) ONs are **bookkeeping**: they assume completely ionic bonds. C–H (ΔEN 0.35) is barely polar, so the CH_3 carbon does not carry a −3 charge next to a −1 one; actual partial charges are small.

Problem 12: Mg + HCl, back-titration (20)

(a) **$Mg(s) + 2\ HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$**; net: **$Mg(s) + 2\ H_3O^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g) + 2\ H_2O(l)$**.
(b) Mg = 0.200/24.305 = 8.229×10^{-3} mol (÷1); HCl = 0.0250 × 0.800 = 0.0200 mol (÷2 = 0.0100) → **Mg limiting**.
(c) H_2 = 8.229×10^{-3} mol; P = 755/760 = 0.9934 atm; V = (8.229×10^{-3})(0.08206)(298.15)/0.9934 = **0.203 L (203 mL)**.
(d) HCl used = 2 × 8.229×10^{-3} = 0.01646 mol → left = 0.0200 − 0.01646 = 3.54×10^{-3} mol → NaOH (1:1) = 3.54×10^{-3}/0.250 M = **14.2 mL**.
Back-titration logic: total acid − acid used = acid titrated.

Problem 13: Uranium enrichment (15)

(a) $\mathcal{M}(^{235}UF_6)$ = 235.044 + 6(18.998) = 349.03; $\mathcal{M}(^{238}UF_6)$ = 352.04 g/mol. $rate_{235}/rate_{238}$ = √(352.04/349.03) = **1.0043** (only 0.43% faster!).
(b) $Ratio_0$ = 0.720/99.280 = 0.007252; target = 3.00/97.00 = 0.03093. $(1.0043)^n$ = 0.03093/0.007252 = 4.265 → n = ln 4.265/ln 1.0043 = **≈ 338 stages**. Use the $^{235}U/^{238}U$ ratio, not the %.
(c) u_rms = √(3RT/$\mathcal{M}$) = √(3 × 8.314 × 333.15/0.35204 kg/mol) = **154 m/s**.

Problem 14: Natural-gas mixture (15)

(a) $\mathcal{M}$_avg = dRT/P = (0.818)(0.08206)(298.15)/1.00 = **20.0 g/mol**.
(b) 16.043χ + 30.069(1 − χ) = 20.01 → χ(CH_4) = (30.069 − 20.01)/14.026 = **0.717**; χ(C_2H_6) = **0.283**.
(c) P_i = χ_i P_tot: **P(CH_4) = 0.717 atm, P(C_2H_6) = 0.283 atm**.
(d) **Ethane**: it is larger (more self-volume) and has stronger intermolecular attractions (higher b.p., −88.5 °C vs −161.5 °C for CH_4).

Problem 15: Safety (10)

(a) **Blue = health, red = flammability, yellow = instability/reactivity, white = special hazards** (W = water-reactive, OX = oxidizer). Scale 0–4: **4 = most severe** (e.g. 4 in red = extremely flammable).
(b) **Procedures, hazards, controls, training documentation**.
(c) **Section 4** (first-aid measures).
(d) Any two: stay alert even in routine work; make safety a daily habit; review lab safety regularly; learn from your own and others' accidents and near-misses.